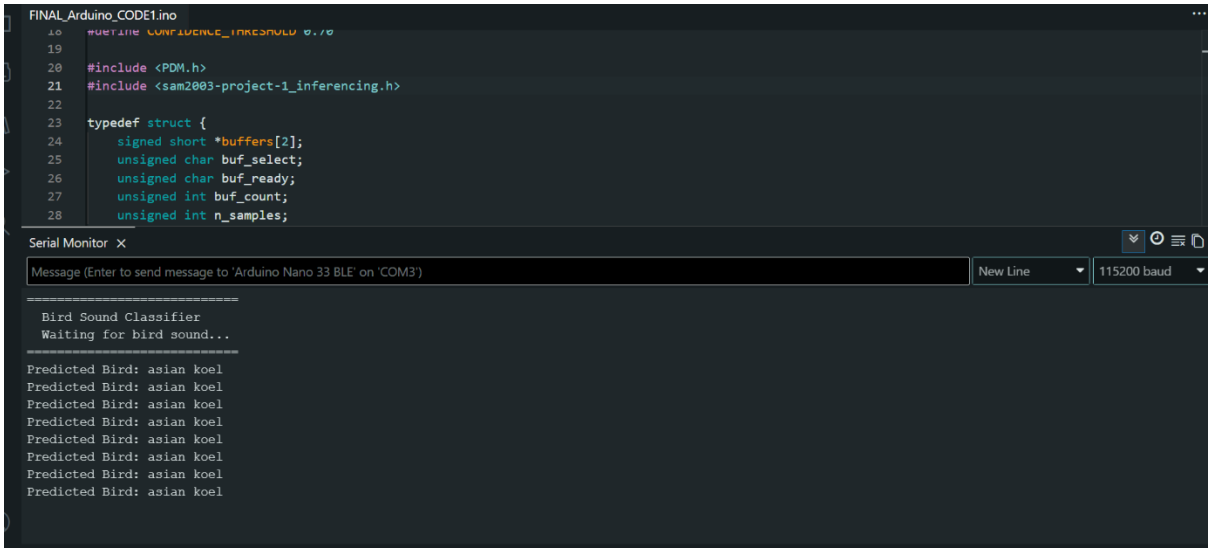
RESULTS

The final testing phase evaluated the complete hardware-to-cloud pipeline during live acoustic events. The system workflow operates in three distinct physical stages to ensure stability. First, the Arduino Nano 33 BLE acts as the localized prediction engine, running the deep learning model offline to identify the avian species. Second, upon a positive identification, the resulting text string is pushed across the RX and TX hardware serial wires to the ESP8266 module. Finally, the ESP8266 acts as a network bridge, utilizing the local Wi-Fi to transmit that text payload directly to the Supabase database for permanent storage.

Live Edge Prediction (Arduino Nano 33 BLE)

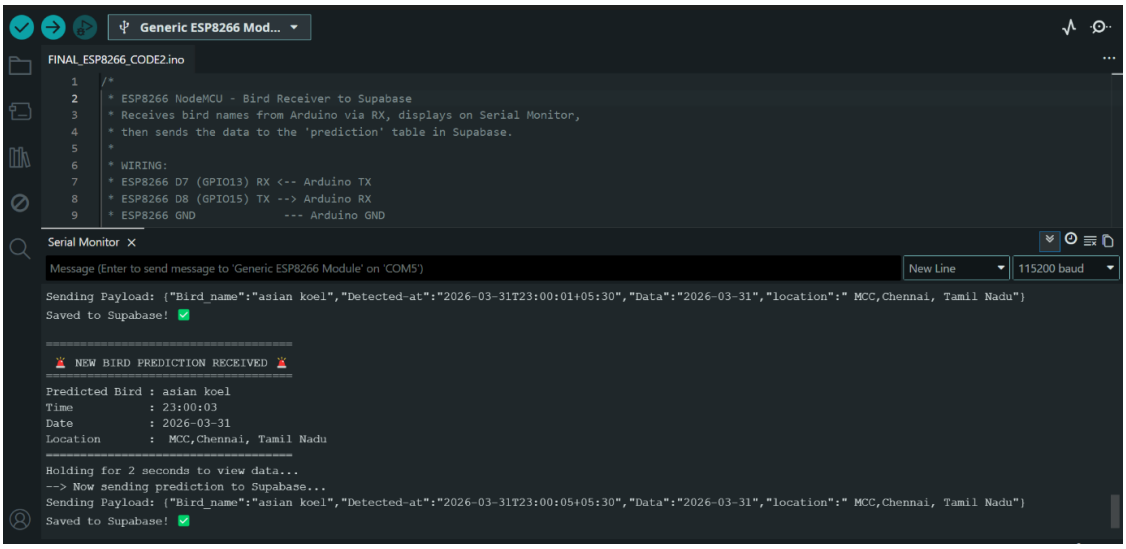
During field deployment, the primary microcontroller successfully handled the continuous audio buffering and model inference without system failures. When a target bird vocalized near the sensor, the embedded Convolutional Neural Network processed the acoustic features and identified the specific pattern. the Arduino immediatel output the correct species name to the local serial port. This successful edge classification verified that the compressed model functions accurately in a real-world environment, isolating the target audio from background wind and campus noise.



Live Edge Prediction (Arduino Nano 33 BLE)

Hardware Serial Routing (ESP8266 Module)

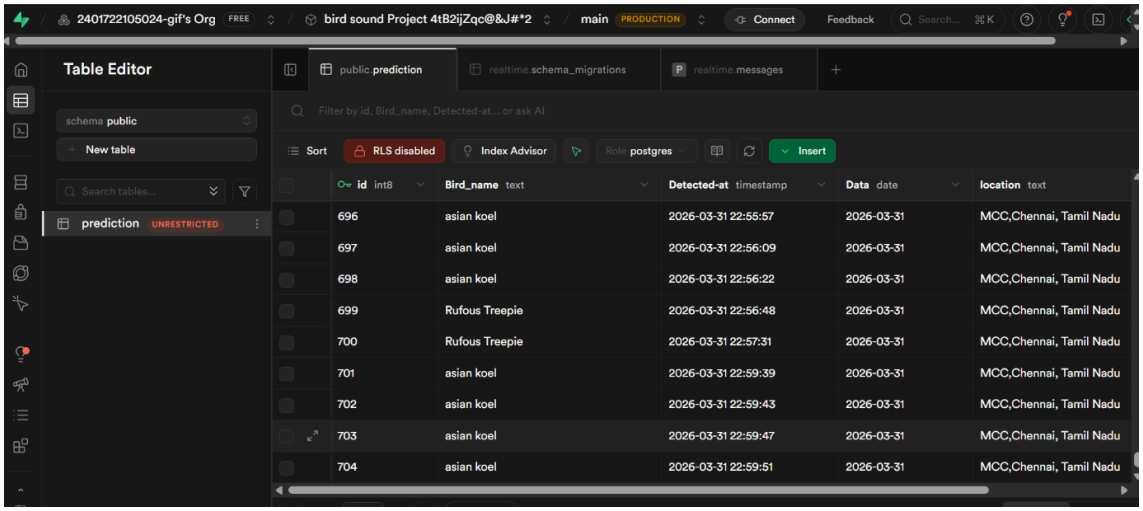
The intermediate communication bridge performed exactly as designed. The ESP8266 maintained a persistent connection to the local internet router while continuously monitoring its hardware receive (RX) pin.the moment the plain text data arrived from the Arduino's transmit (TX) pin, the ESP8266 packaged the string into a structured JSON format. Because the physical workload was divided, the ESP8266 was able to focus entirely on formatting the HTTP POST request. This ensured the transmission occurred instantly without dropping data packets or crashing the local listening loop.



Hardware Serial Routing (ESP8266 Module)

Remote Database Logging (Supabase Platform)

The final validation of the system was confirming the cloud storage pipeline. By monitoring the Supabase backend during the live outdoor tests, the incoming data was verified in real-time. whenever the ESP8266 transmitted a payload over the Wi-Fi network, a new row automatically populated within the project database. The platform successfully captured the plain text bird name and immediately applied an accurate, automated server-side timestamp to the event. This resulted in a clean, chronological digital log of campus wildlife activity that required zero manual human intervention.



Remote Database Logging (Supabase Platform)

Circuit Design and Wiring

